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4 DECEMBER 1951

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M3 FOR 81-MM MORTAR AMMUNITION

This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers two classes of M3 Ignition Cartridges for use in 81-mm mortar ammunition.

Class A — Primed, and loaded.
Class B — Primed, but not loaded.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 — General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-129— Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

75-19-76 — Cartridge, Ignition, M3, for 81-mm Mortar, M1 Ammunition.
81-3-44 — Fixture, Primer Sensitivity, Testing
81-3-46 — Fixture, Primer Sensitivity, Testing

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M12-4 — Cartridge, Ignition, M3 and M4, Loaded.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.**—Material and parts shall be in accordance with applicable specifications and drawings.

3.2 The ignition cartridges shall comply with all requirements specified on Drawing 75-19-76 and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Primer mixture.

3.3.1 *Specified primer mixture.*—The total percentage of the ingredients specified on the drawing for use in the primer composition shall be not less than 95 percent when tested as specified in 4.4.

3.3.2 *Special primer mixture.*—In case the contractor desires to use a primer mixture other than the mixture specified on the drawing, he shall furnish a signed statement showing the ingredients which he proposes

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to use in the mixture, the purity of or specifications for the ingredients and the percentage of each with the permissible tolerances, together with a method of analysis of the mixture and of the individual ingredients in case the ingredients should be new or original materials, for the approval of the United States Government prior to the use thereof. Said approval shall be in writing.

3.4 Bursting strength.—The ignition cartridge shall have a maximum bursting strength of 141 pounds and a minimum bursting strength of 55 pounds when tested as specified in 4.6.

3.5 Sensitivity.

3.5.1 *Nonfunctioning.*—The ignition cartridges shall not function under the impact of a 2-ounce ±0.2 weight dropped from a height of 2 inches when tested as specified in 4.7.1.

3.5.2 *Functioning.*

3.5.2.1 *Functioning on center.*—The ignition cartridge shall function, when struck on center, under the impact of a 2-ounce ±0.2 weight dropped from a height of 12 inches when tested as specified in 4.7.2.1.

3.5.2.2 *Functioning off center.*—The ignition cartridge shall function when struck 0.02 inches off center under the impact of a 2-ounce ±0.2 weight dropped from a height of 12 inches when tested as specified in 4.7.2.2.

3.6 Proving ground requirements for the ignition cartridge.

3.6.1 All cartridges shall function when tested as specified in 4.8.1.

3.6.2 The average muzzle velocity shall be 235 ±10 feet per second when tested as specified in 4.8.1.

3.6.3 The difference between the maximum and minimum muzzle velocities shall be not greater than 20 feet per second when tested as specified in 4.8.1.

3.6.4 All cartridges shall leave the bore of the mortar with the projectile when tested as specified in 4.8.1.

3.6.5 Not more than five of the sample cartridges fired shall leave any part of the primer in the bore of the mortar when tested as specified in 4.8.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.—A lot shall consist of ignition cartridges loaded and assembled by one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, the same drawing revision, same specification, and same specification revision. Each lot shall contain:

(a) Ignition cartridges of one class.
(b) Propellant from not more than one lot.

4.2 Sampling. — Number of samples and acceptance and rejection criteria used for determination of lot acceptance for test as specified in 4.6 and 4.7 shall be in accordance with Standard Inspection Procedure ORD–SIP–M12–4. Fifty sample ignition cartridges shall be taken from each lot by the inspector for proving ground test as specified in 4.8. Sampling of component materials is specified in paragraphs covering test of the materials.

4.3 Inspection. — Inspection shall be as specified in Specification MIL–G–2550.

4.4 Determination of specified primer mixture.—The moisture content of the primer mixture shall be determined and the ingredients, as specified on the drawing shall be verified at intervals as specified by the inspector. When such tests are made, the primer mixture shall be selected before loading or removed from fifty primers and analyzed as follows to determine compliance with 3.3.1.

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4.4.1 *Moisture.*—A known weight, 0.40 to 0.70 gram of the primer mixture shall be dried to constant weight at 55° to 65° C., or in a desiccator containing calcium chloride. The loss in weight shall be calculated as the percentage of moisture.

4.4.2 *Composition.*

4.4.2.1 *Mercury fulminate.* — The amount of sample to be used will vary with the composition. An amount of sample which will contain approximately 0.3 gram of mercury fulminate accurately weighted shall be placed in a dry 250-ml beaker moistened with 1 to 2 drops of alcohol and at room temperature and 30 ml of 20-percent sodium-thiosulphate solution shall be added. The solution shall be stirred thoroughly for 1 minute, diluted to about 100 ml with distilled water, and titrated with N/10 sulphuric acid, using methyl red as indicator. It is important to begin the titration exactly 1 minute after the addition of the thiosulphate solution, and to run the acid from the burette at such a rate that the time of titration does not exceed 1 minute. A blank determination shall be made using the same amount of sodium thiosulphate solution, alcohol, and water, and making the necessary corrections.

$$\text{Percent mercury fulminate} = \frac{(A-B)\times N\times 0.0711565\times 100}{S}$$

Where:

A = ml of acid used in titration.
B = ml of acid required for blank determination.
N = normality of acid.
S = weight of sample.

4.4.2.2 *Antimony sulfide.* — The contents of the beaker remaining at the end of the mercury-fulminate titration shall be filtered immediately through an ignited, tared filtering crucible. The insoluble material shall be washed with water at room temperature; the filtrate and washings transferred to a 250-ml beaker; the residue washed with small amounts of alcohol and ether and dried at 100° C. If unknown, the identity of the residue as antimony sulfide or a mixture of antimony sulfide and glass shall be determined by microscopic or other examination of the residue from an extra sample. The crucible containing the regular sample shall be cooled and weighted. If antimony sulfide only is present, the increase in weight of the crucible shall be calculated as the percentage of antimony trisulfide in the sample. In the presence of both antimony sulfide and ground glass, the bulk mixture shall be transferred to a small beaker, taking care not to disturb the asbestos mat; 40 ml of concentrated hydrochloric acid and 10 ml of concentrated nitric acid shall be added and boiled until the solution of antimony sulfide is complete. The solution shall be filtered through the filtering crucible, taking care to dissolve any antimony sulfide which has remained in the crucible. The solution shall be ignited to a dull red heat, cooled in a desiccator, and weighed. The loss in weight shall be calculated as the percentage of antimony trisulfide in the sample.

4.4.2.3 *Ground glass.*—The original weight of the filtering crucible shall be substracted from its weight after the acid extraction in the antimony sulfide determination and the difference calculated as the percentage of ground glass in the sample.

4.4.2.4 *Potassium chlorate.* — One ml of concentrated sulfuric acid and 20 ml of 10-percent ferrous-ammonium-sulfate solution shall be added to the filtrate and waterwashings from the antimony trisulfide determination and then boiled for 10 minutes. The precipitated mercuric sulfide shall be filtered off; 2 ml of concentrated nitric acid shall be added; and the solution boiled for 5 minutes or more. The solution shall be cooled then precipitated with 5 ml of 10-percent silver-nitrate solution. The precipitate shall be allowed to settle overnight; then filtered through a tared filtering crucible; washed

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with water; then the water shall be removed with two alcohol washes. The solution shall be washed with C. P. carbon disulfide until all the sulfur has been removed; dried at 130° C.; cooled and weighed.

$$\text{Percent potassium chlorate} = \frac{W \times 0.8550 \times 100}{S}$$

Where:

W = Weight of silver chloride obtained.
S = Weight of sample in grams.

4.4.2.5 *Water-soluble binder.*—The water soluble binder shall be determined by difference; that is, by subtracting from 100 percent the sum of all the percentages previously determined.

4.5 Test of special mixture. — In case a special primer mixture is used as provided in 3.3.2, the moisture content of the primer mixture shall be determined and the ingredients shall be verified at intervals as prescribed by the inspector. When such tests are made, the primer mixture shall be selected before loading or removed from 50 primers and analyzed according to the method approved by the contracting officer to determine compliance with 3.3.2.

4.6 Determination of bursting strength. — Sample ignition cartridges shall be selected, and powder charge removed by cutting off the crimp. The body assembly shall be slit lengthwise as shown on figure 1 and one-half placed in a fixture (see fig. 2). The bursting strength shall be determined in a machine satisfactory to the inspector. The load shall be applied at the rate of .100 to .125 inch per minute, inclusive to determine compliance with 3.4.

4.7 Sensitivity test.

4.7.1 *Nonfunctioning.*—The powder charge shall be removed from each sample class A cartridge before testing. Each cartridge shall be inserted in an adaptor ring (see fig. 3) and the adapter placed in the base of the primer sensitivity testing fixture, Drawing 81–3–44. Firing pin (see figure 4) shall be inserted in type B body, Drawing 81–3–46, and placed into weight cup support so that firing pin firmly contacts the primer, as shown by assembly. The height of drop of a steel ball (see 6.1) weighing 2 ±0.2 ounces shall be adjusted to 2 inches. The ball shall be released. Observe for nonfunctioning to determine compliance with 3.5.1.

4.7.2 *Functioning.*

4.7.2.1 *Functioning on center.*—The sample cartridges shall be subjected to the test specified in 4.7.1, except that the height of drop shall be 12 inches. Observe for functioning on center to determine compliance with 3.5.2.1.

4.7.2.2 *Functioning off center.*—The sample cartridges shall be subjected to the test specified in 4.7.1, except that the adapter ring (see fig. 3) shall be so adjusted in the primer sensitivity testing fixtures, Drawing 81–3–44, that the firing pin holder contacts the upper face of the adapter ring at 0.02 inch off center. Observe for functioning off center to determine compliance with 3.5.2.2.

4.8 Proving ground test.

4.8.1 *Procedure.*

4.8.1.1 *Class A.*—Twenty-five sample cartridges shall be fired in an 81-mm mortar using an inert M52 Fuze and an M43A1 Practice Shell loaded to a total sand weight of 7.04 pounds. (If necessary add shot to obtain the desired weight.) The mortar shall be elevated to an angle of 45°. The velocity shall be measured, and observation shall be made for functioning, to see that the cartridge leaves the bore of the mortar with the projectile or if any part of the primer remains in the mortar, to determine compliance with 3.6.

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4.8.1.2 *Class B.*—Twenty-five sample cartridges shall be loaded with the powder charge specified on the drawing for the primed and loaded cartridges. The sample cartridges shall be fired in an 81-mm mortar using an inert M52 Fuze and an M43A1 Practice Shell loaded to a total weight of 7.04 pounds. (If necessary add shot to obtain the desired weight.) The mortar shall be elevated to an angle of 45°. The velocity shall be measured, and observation shall be made for functioning, to see that the cartridge leaves the bore of the mortar with the projectile or if any part of the primer remains in the mortar, to determine compliance with 3.6.

4.8.2 *Acceptance.*—The lot shall be considered acceptable provided that in the phase being tested (i.e., primed and loaded or primed, but not loaded) the cartridges functioned and complied with 3.6.1, 3.6.2, 3.6.3, 3.6.4, and 3.6.5. The lot shall be considered eligible for retest under the provisions of 4.8.3, provided:

(a) Not more than two of the sample cartridges fail to function, or

(b) The average muzzle velocity is less than 225 feet per second, or greater than 235 feet per second, or

(c) The difference between the maximum and minimum muzzle velocities is greater than 20 feet per second, or

(d) Not more than two of the sample cartridges fail to leave the bore of the mortar with the projectiles, or

(e) Not more than eight of the sample cartridges fired leave any part of the primer in the bore of the mortar.

The lot shall be rejected if the limits for eligibility for retest are exceeded.

4.8.3 *Retest.* — A retest as specified in 4.8.2 may be made, if requested by the contractor, using double the number of samples used in the original test. The lot shall be considered acceptable on retest provided that in the phase being retested (i.e., primed and loaded, primed, but not loaded) all the sample cartridges comply with 3.6.1, 3.6.2, 3.6.3, 3.6.4 and 3.6.5.

5. PREPARATION FOR DELIVERY

5.1 Packing.—Unless otherwise specified, the cartridges shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

5.2 Marking.—Marking shall be in accordance with Standard MIL–STD–129.

5.3 Data card. — Data card information shall be as specified in Specification MIL–G–2550.

6. NOTES

6.1 Commercial steel ball bearings can be purchased, which will comply with the requirements for steel balls used in the sensitivity tests.

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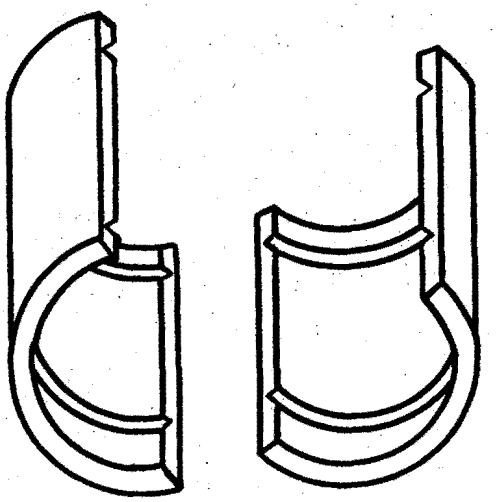


FIGURE 1.

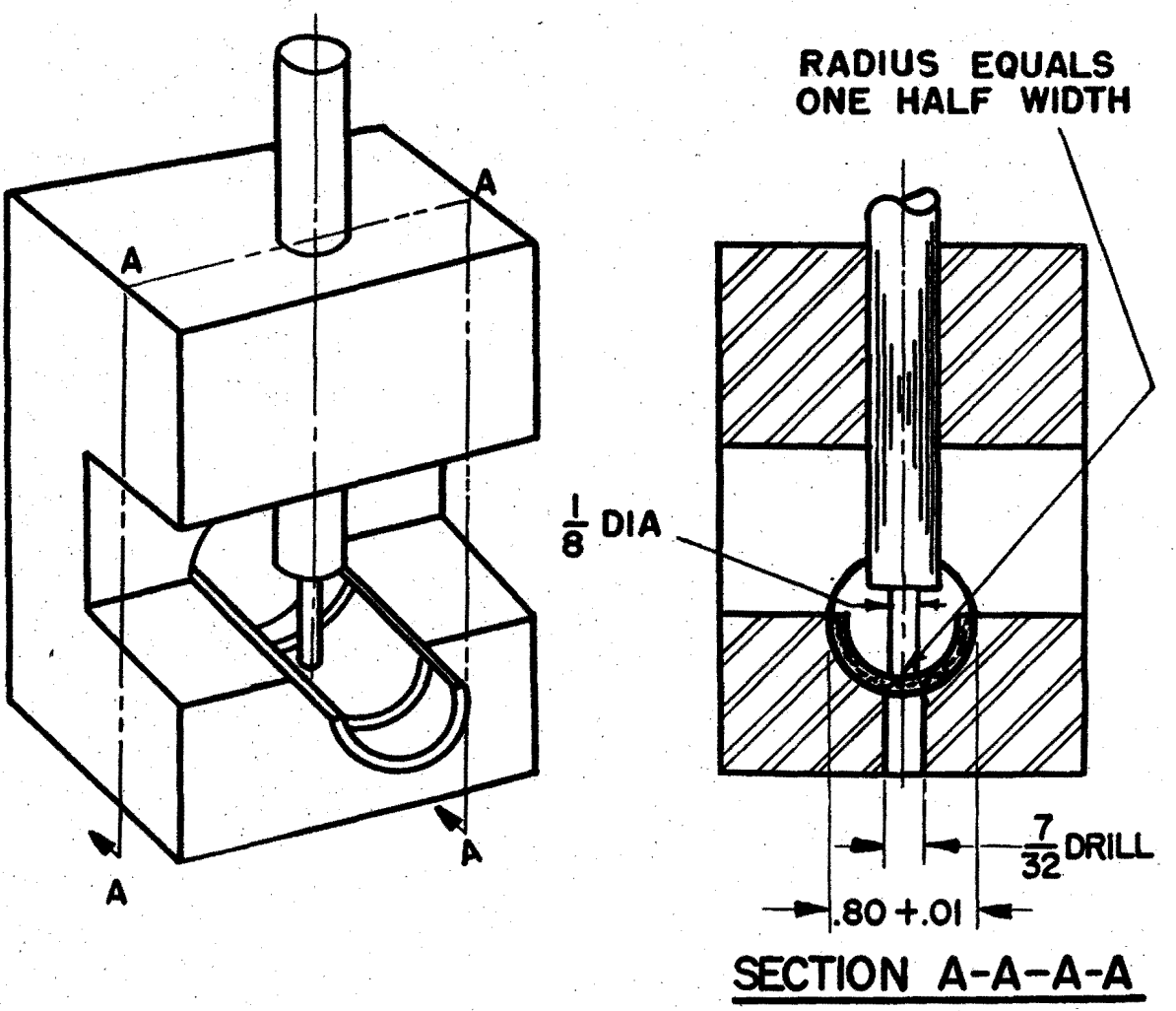


FIGURE 2.

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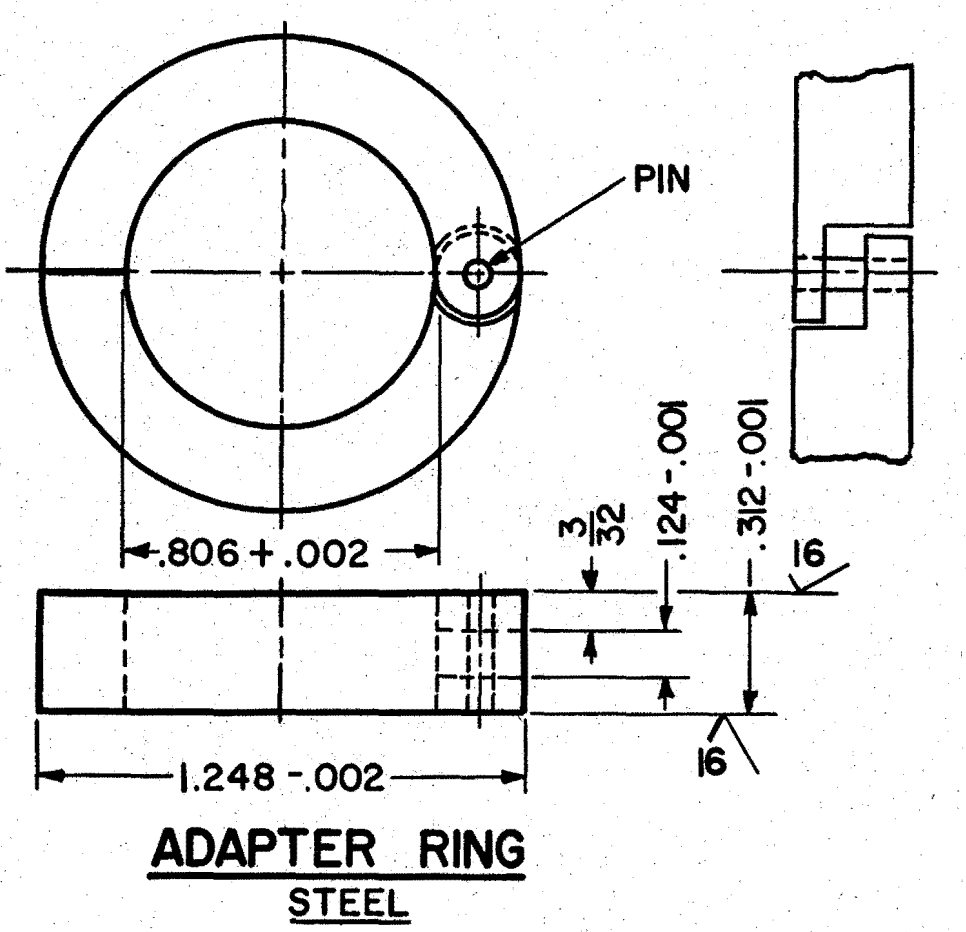


FIGURE 3. *Adapter ring, steel.*

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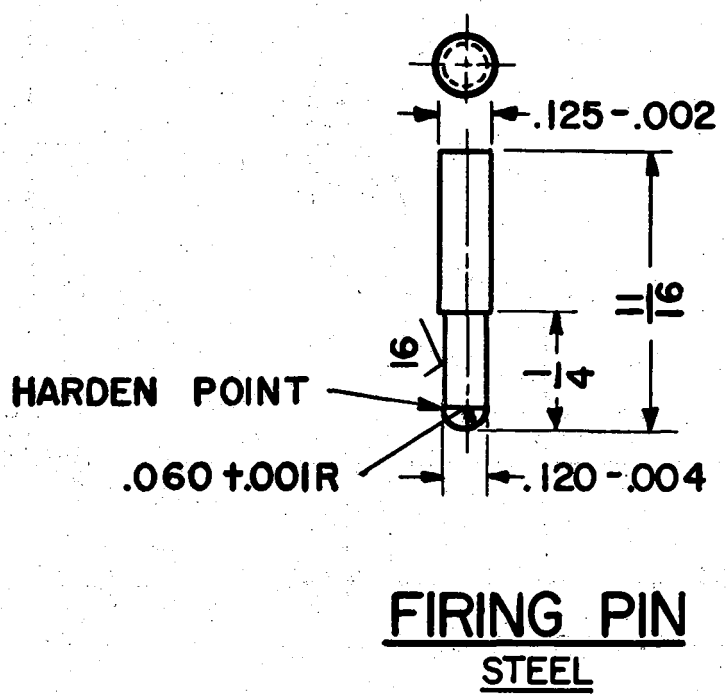


FIGURE 4. *Firing pin, steel.*

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